

Walkthrough: Automated Test Suite Implementation & Execution

We have implemented and executed an automated end-to-end API test suite for the **Birhane Hiwot Sunday School Attendance System** (ብርሃነ ሕይወት ሰንበት ትምህርት ቤት) based on [docs/qa/TEST-CASES.md] (file:///c:/Users/Ym/Desktop/b/ss/Birhane-Hiwot-sunday-school-attendance/docs/qa/TEST-CASES.md).

1. Automated Test Suite Highlights

Script: [scripts/run-automation-tests.mjs](file:///c:/Users/Ym/Desktop/b/ss/Birhane-Hiwot-sunday-school-attendance/scripts/run-automation-tests.mjs)

- **Native ESM Runner:** Pure Node.js test runner with colored terminal outputs, timing metrics, and non-destructive execution with automatic teardown.
- **Session & Cookie Jar Management:** Handles CSRF tokens, NextAuth credentials callbacks, `next-auth.session-token` cookies, and Mobile JWT auth tokens.
- **Added Missing Endpoint:** Implemented [src/app/api/facilitators/roles/route.ts] (file:///c:/Users/Ym/Desktop/b/ss/Birhane-Hiwot-sunday-school-attendance/src/app/api/facilitators/roles/route.ts) returning canonical facilitator roles (`Attendance Facilitator` and `Education Facilitator`).
- **NPM Integration:** Added `"test:api"` and `"test"` commands in [package.json] (file:///c:/Users/Ym/Desktop/b/ss/Birhane-Hiwot-sunday-school-attendance/package.json).

2. Test Execution Results

Command executed: `npm run test (node scripts/run-automation-tests.mjs)`

```
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Birhane Hiwot Sunday School – Automated QA Test Suite
(ብርሃነ ሕይወት ሰንበት ትምህርት ቤት የአቴንዳንስ እና የትምህርት ክፍል)
Target: http://localhost:3000
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► Suite 1: Authentication, Setup & Mobile JWT
  ✓ PASS [AUTH-012] Super Admin Setup (one-time bootstrap / verification)
(403.5ms)
  ✓ PASS [AUTH-013] Setup with invalid confirm token fails with 400 (223.1ms)
  ✓ PASS [MOB-001] Mobile JWT authentication returns token & user payload
(567.3ms)
  ✓ PASS [MOB-002] Mobile authentication fails on invalid password (401)
(723.1ms)
  ✓ PASS [MOB-003] Mobile auth OPTIONS CORS preflight returns 204 (200.7ms)
  ✓ PASS [AUTH-011] NextAuth CSRF token retrieval (549.3ms)
  ✓ PASS [AUTH-001] NextAuth credentials login callback (944.2ms)
  ✓ PASS [AUTH-010] Verify active web session (397.7ms)
```

► Suite 2: Facilitators & Roles Management

- ✓ PASS [FAC-011] List facilitator roles constants (379.6ms)
- ✓ PASS [FAC-003] Create Attendance Facilitator with assigned grade (1158.1ms)
- ✓ PASS [FAC-004] Create Attendance Facilitator without grade fails with 400 (471.1ms)
- ✓ PASS [FAC-001] List facilitators (362.1ms)
- ✓ PASS [FAC-009] Toggle canAddStudent permission via PATCH (395.4ms)
- ✓ PASS [FAC-006] Update facilitator profile via PUT (1022.6ms)
- ✓ PASS [FAC-010] Get total facilitators count (756.8ms)
- ✓ PASS [FAC-012] Education Facilitator CRUD flow (1923.0ms)

► Suite 3: Teachers, Subjects & Curriculum

- ✓ PASS [TCH-002] Create Teacher with assigned subjects & grade (605.8ms)
- ✓ PASS [TCH-001] List teachers (262.8ms)
- ✓ PASS [SUB-002] Add subject to grade group (437.7ms)
- ✓ PASS [SUB-001] List grouped subjects for academic year & grade (288.2ms)
- ✓ PASS [SUB-004] Batch create authentic Orthodox Sunday School subjects (319.9ms)
- ✓ PASS [TAS-002] Create teacher-to-subject assignment (879.0ms)
- ✓ PASS [TAS-001] List teacher assignments (350.0ms)

► Suite 4: Category Registration Periods

- ✓ PASS [PER-001] List category periods (411.2ms)
- ✓ PASS [PER-003] Create / Upsert Category Period for Regular classification (304.6ms)
- ✓ PASS [PER-004] Category Period rejects missing fields with 400 (211.1ms)

► Suite 5: Student Registration & Profile Management

- ✓ PASS [STU-006] Register new student with authentic Amharic Ethiopian data (737.2ms)
- ✓ PASS [STU-007] Student registration missing required field returns 400 (165.0ms)
- ✓ PASS [STU-020] Check duplicate student returns true for existing student (326.4ms)
- ✓ PASS [STU-021] Check duplicate student returns false for unrecorded student (258.3ms)
- ✓ PASS [STU-001] List students with grade and academic year filter (254.1ms)
- ✓ PASS [STU-004] Get student by Unique_ID query param (235.9ms)
- ✓ PASS [STU-005] Get student by ID or Unique_ID path param (299.9ms)
- ✓ PASS [STU-023] Count students by grade and academic year (362.1ms)
- ✓ PASS [STU-025] Get total registered students count (339.1ms)
- ✓ PASS [STU-015] Update student profile via PUT (416.4ms)
- ✓ PASS [STU-017] Update student photo via PATCH (375.0ms)
- ✓ PASS [STU-018] Regenerate student QR code via PATCH (572.1ms)

► Suite 6: Bulk Attendance & Summaries

- ✓ PASS [ATT-006] Mark attendance (bulk upsert) (612.0ms)
- ✓ PASS [ATT-009] Idempotent attendance re-submit updates without duplicate rows (299.2ms)
- ✓ PASS [ATT-001] List attendance records by date (409.6ms)
- ✓ PASS [ATT-003] Attendance summary mode returns total & present counts (371.6ms)
- ✓ PASS [ATT-012] Get student attendance history (301.0ms)

- ▶ Suite 7: Academic Results & University Letter Grading
 - ✓ PASS [RES-003] Record student result and verify letter grade calculation (A+) (597.4ms)
 - ✓ PASS [RES-001] List all student results (301.4ms)
 - ✓ PASS [RES-002] Filter student results by studentId (340.5ms)
 - ✓ PASS [RES-010] Get student result by ID (333.5ms)
 - ✓ PASS [RES-011] Update student result recalculates total & grade (436.5ms)
- ▶ Suite 8: 13-Month Ethiopian Payment Tracking
 - ✓ PASS [PAY-001] Lazy initialization of all 13 Ethiopian months (754.8ms)
 - ✓ PASS [PAY-003] Save monthly fee payment status (278.1ms)
- ▶ Suite 9: Student Requests Approval Workflow
 - ✓ PASS [REQ-001] Submit student registration request (494.0ms)
 - ✓ PASS [REQ-003] List pending student requests (496.6ms)
 - ✓ PASS [REQ-004] Get student request by ID (484.3ms)
 - ✓ PASS [REQ-005] Approve student request (405.2ms)
- ▶ Suite 10: QR Code Verification & System Utilities
 - ✓ PASS [QR-005] QR verify rejects empty payload with 400 (328.6ms)
 - ✓ PASS [UTIL-001] Google Sheets health check endpoint (2529.0ms)
 - ✓ PASS [UTIL-002] Attendance aggregation cron endpoint (419.2ms)
- ▶ Cleanup: Removing Temporary Automated Test Data
 - ✓ PASS [TEARDOWN-001] Delete created test student (312.1ms)
 - ✓ PASS [TEARDOWN-002] Delete created test student result (311.8ms)
 - ✓ PASS [TEARDOWN-003] Delete created teacher assignment (2858.7ms)
 - ✓ PASS [TEARDOWN-004] Remove test subject from group (468.4ms)
 - ✓ PASS [TEARDOWN-005] Delete created test teacher (552.6ms)
 - ✓ PASS [TEARDOWN-006] Delete created test facilitator (507.2ms)

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Automated Test Execution Summary

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Total Tests Run : 63
Passed : 63
Failed : 0
Pass Rate : 100.0%

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3. How to Re-Run Anytime

You can execute the automated test suite at any time via:

```
npm run test
# or
npm run test:api
```